The Transformations Bistro

Assessment Menu · choose your meal and show what you know

Name: _____

Date: _____

How to order

1. Order **4 Appetizers** (5 points each), **4 Entrées** (10 points each) and **2 Chef's Specials** (20 points each). The most you can earn is **100 points**.
2. Your 10 dishes must include **at least one of each skill**: PLANE, SHIFT, SPIN and COMBINE (the label next to each dish).
3. Mark your choices on the order form below **and** on the dishes themselves ("My pick"). Only marked dishes are graded, and you cannot swap a dish after turning it in.
4. Show your work in the space for each dish. Points are given for correct steps as well as correct answers.
5. Appetizers and Entrées are on pages 2 and 3. The Chef's Specials are on page 4. Look at every page before you choose!

Appetizers

5 pts each · choose 4

- ☐ **A1** PLANE
- ☐ **A2** PLANE
- ☐ **A3** SHIFT
- ☐ **A4** SHIFT
- ☐ **A5** SPIN
- ☐ **A6** COMBINE

Entrées

10 pts each · choose 4

- ☐ **E1** PLANE
- ☐ **E2** SHIFT
- ☐ **E3** SPIN
- ☐ **E4** SPIN
- ☐ **E5** COMBINE
- ☐ **E6** COMBINE

Chef's Specials

20 pts each · choose 2

- ☐ **S1** PLANE
- ☐ **S2** SHIFT
- ☐ **S3** SPIN
- ☐ **S4** COMBINE

My skill check (circle each skill you have covered): PLANE · SHIFT · SPIN · COMBINE

Course	Dishes	Points each	Possible	Score (teacher)
Appetizers	4	5	20	
Entrées	4	10	40	
Chef's Specials	2	20	40	
Total			**100**	

Appetizers

5 points each · choose 4 · Warm-ups

Name: _____

A1 PLANE ☐ My pick · 5 pts

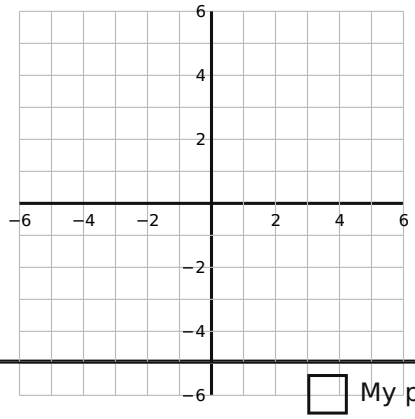
Find the distance between A(1, 2) and B(4, 6).

A2 PLANE ☐ My pick · 5 pts

Find the midpoint and the slope of the segment with endpoints A(−2, 5) and B(6, −1).

A3 SHIFT ☐ My pick · 5 pts

Triangle PQR has P(−3, 2), Q(1, 4), R(0, −1). Translate it using (x, y) → (x + 4, y − 3). Graph the image and label P′, Q′, R′.



A4 SHIFT ☐ My pick · 5 pts

Point A is at (3, −5). Give the coordinates of its reflection across the x-axis and its reflection across the y-axis.

A5 SPIN ☐ My pick · 5 pts

Rotate the point (2, 5) by 90° counterclockwise about the origin. Use the rule (x, y) → (−y, x).

A6 COMBINE ☐ My pick · 5 pts

Point (−2, 3) is reflected across the y-axis, then translated 5 units down. Find its final image.

Entrées

10 points each · choose 4 · Multi-step problems

Name: _____

E1 PLANE ☐ My pick · 10 pts

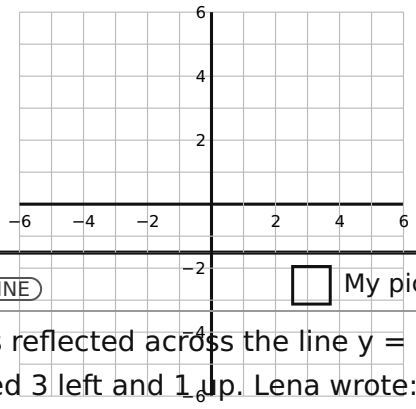
Triangle ABC has A(−1, 1), B(3, 4), C(6, 0). **(a)** Use the distance formula to find AB, BC and AC. **(b)** Use slopes to decide whether ∠B is a right angle. **(c)** Classify the triangle by its sides.

E2 SHIFT ☐ My pick · 10 pts

A translation maps A(−2, 3) to A′(4, −1). **(a)** Write the translation rule. **(b)** Find the image of B(1, −5). **(c)** Which point is mapped to (0, 0)?

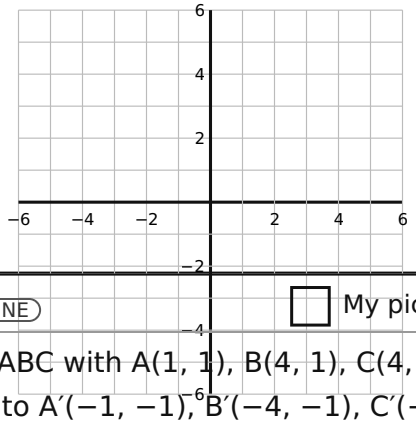
E3 SPIN ☐ My pick · 10 pts

Triangle ABC has A(2, −4), B(6, −2), C(4, 4). Dilate it with center at the origin and scale factor $\frac{1}{2}$. Graph both triangles, list the image points, and compare AB with A′B′ using the distance formula.



E4 SPIN ☐ My pick · 10 pts

Triangle DEF has D(1, 2), E(4, 2), F(4, 5). Rotate it 90° clockwise about the origin using (x, y) → (y, −x). Graph the image and use the distance formula to compare DE with D′E′. What does this show?



E5 COMBINE ☐ My pick · 10 pts

P(2, 5) is reflected across the line y = x, then translated 3 left and 1 up. Lena wrote: “Reflection: (2, 5) → (−5, −2). Translation: (−8, −1).” Find her mistake and give the correct final image.

E6 COMBINE ☐ My pick · 10 pts

Triangle ABC with A(1, 1), B(4, 1), C(4, 3) is mapped to A′(−1, −1), B′(−4, −1), C′(−4, −3). **(a)** Describe the single transformation and write its rule. **(b)** Show that the origin is the midpoint of AA′.

Chef’s Specials

20 points each · choose 2 · Multi-part challenges

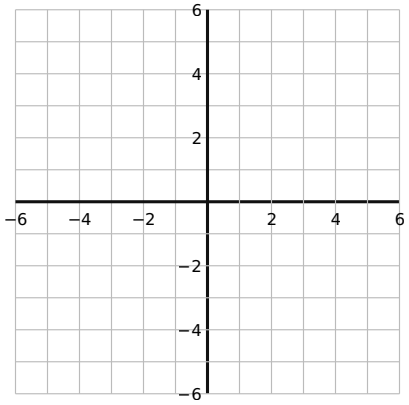
Name: _____

S1 PLANE ☐ My pick · 20 pts

A(−2, 1), B(1, 5), C(7, 3) are three vertices of parallelogram ABCD, in that order. **(a)** Use the fact that diagonals bisect each other to find D. **(b)** Use slopes to verify ABCD is a parallelogram. **(c)** Is it a rectangle? Justify with slopes.

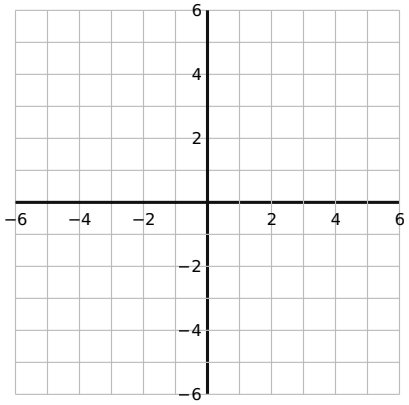
S2 SHIFT ☐ My pick · 20 pts

Triangle LMN has L(2, 1), M(5, 1), N(5, 5). **(a)** Reflect it across the line x = 1. Graph both triangles and the line, and list L′, M′, N′. **(b)** Show the line x = 1 is the perpendicular bisector of MM′ using the midpoint and slopes.



S3 SPIN ☐ My pick · 20 pts

A dilation centered at the origin maps A(3, −6) to A′(1, −2). **(a)** Find the scale factor. **(b)** Find the image of B(−9, 12). **(c)** Find AB and A′B′ in simplest radical form and show their ratio equals the scale factor. **(d)** Triangle ABC has area 54. Find the area of its image and explain.

S4 COMBINE ☐ My pick · 20 pts

Triangle ABC has A(1, 2), B(5, 2), C(5, 4). **(a)** Rotate it 90° counterclockwise about the origin using (x, y) → (−y, x). **(b)** Reflect that image across the x-axis. Graph ABC and the final triangle and list the final points. **(c)** Name the single transformation with the same result. **(d)** If A is reflected first and rotated second, does it land at the same point? Explain.